ELE 734 - Pre-Lab #3

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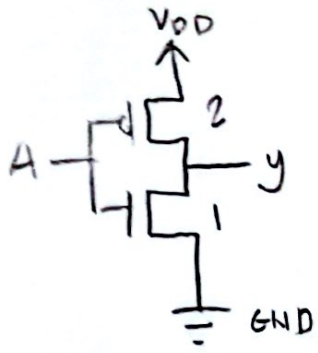
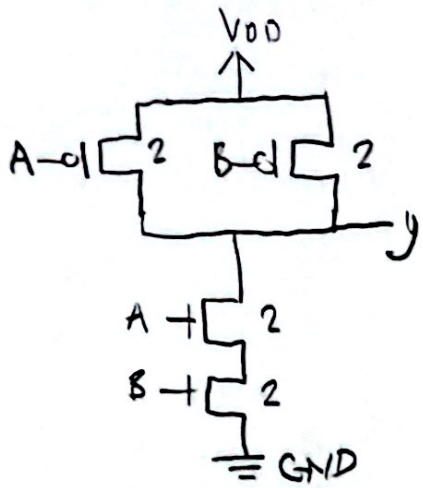
(1)

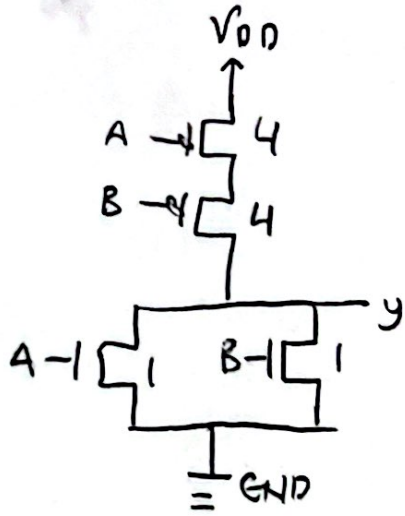
Unskewed:



(Inverter)

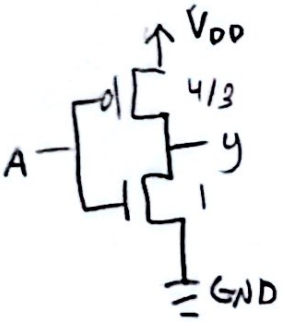


(NAND 2)

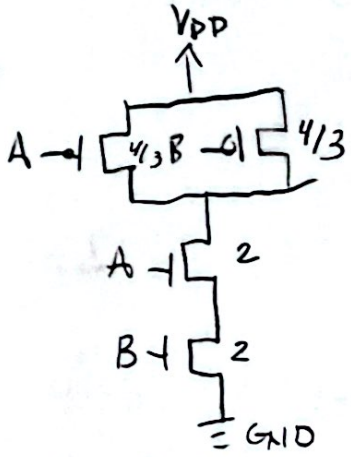


(NOR 2)

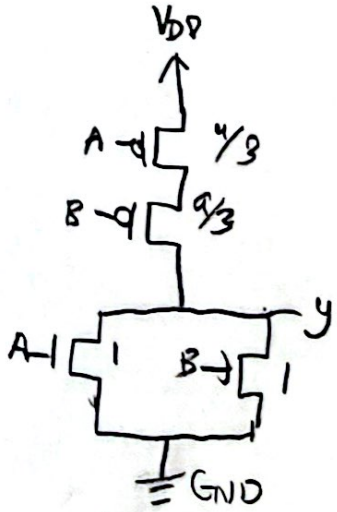
Lo-skewed:



(Inverter)

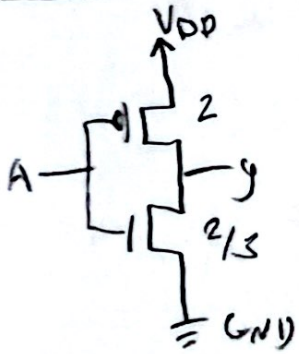


(NAND 2)

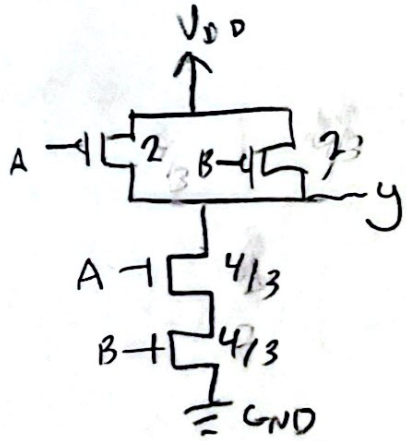


(NOR 2)

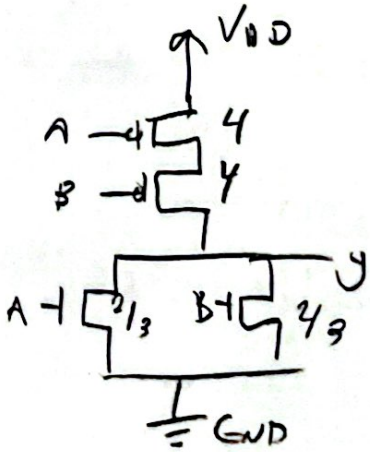
Hi-skewed:



(Inverter)



(NAND 2)



(NOR 2)

Scanned with CamScanner

(2)

NAND 2:

↳ Unskewed-UP:

$g_u = \frac{C_{in}}{C_{inv}} = \frac{2+2}{2+1} = 4/3$

$P_u = \frac{C_{out}}{C_{inv}} = \frac{2+2+2}{2+1} = 2$

↳ Unskewed-down:

$g_d = \frac{C_{in}}{C_{inv}} = \frac{2+2}{2+1} = 4/3$

$P_u = \frac{C_{out}}{C_{inv}} = \frac{2+2+2}{2+1} = 2$

↳ Lo-skewed-Up

$\frac{2R}{\frac{2R}{4/3}} = 4/3$, $g_u = \frac{2+4/3}{4/3+2/3} = 1.67$

↳ Lo-skewed-down

$\frac{R}{\frac{R}{2}\cdot 2} = 1$, $g_d = \frac{2+4/3}{3} = 1.11$

↳ Hi-skewed-up

$g_u = \frac{2+4/3}{3} = 1.11$

↳ Hi-skewed-down

$\frac{R}{\frac{R}{4/3}\cdot 2} = 2/3$, $g_d = \frac{2+4/3}{4/3+2/3} = 1.67$

$g_{avg}(\text{unskewed}) = 4/3 = 1.33$

$g_{avg}(\text{lo-skewed}) = \frac{g_u+g_d}{2} = 1.388$

$g_{avg}(\text{hi-skewed}) = g_{avg} = 1.388$

NOR2:

↳ Unskewed-up

$g_u = \frac{C_{in}}{C_{inv}} = 5/3$

↳ Unskewed-down

$g_d = \frac{C_{in}}{C_{inv}} = 5/8$

↳ Lo-skewed-up

$\frac{2R}{\frac{2R}{8/3}\cdot 2} = 4/3$, $g_u = \frac{8/3+1}{4/3+2/3} = 1.833$

↳ Lo-skewed-down

$g_d = \frac{8/3+1}{3} = 1.22$

↳ Hi-skewed-up

$g_u = \frac{2/3+4}{3} \approx 1.556$

↳ Hi-skewed-down

$\frac{R}{\frac{R}{2/3}} = 2/3$, $g_d = \frac{2+4/3}{4/3+2/3} = 2.33$

$g_{avg}(\text{unskewed}) = 5/3 = 1.667$

$g_{avg}(\text{lo-skewed}) = \frac{g_u+g_d}{2} = 1.5275$

$g_{avg}(\text{hi-skewed}) = g_{avg} = 1.94465$

Scanned with CamScanner